

CS 7545 F26, Homework 1

Zejia Chen
GTID: 904178159

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Exercise 1

Fix $i \neq j$. By independence of entries of $\vec{x}^{(i)}$ and $\vec{x}^{(j)}$, we have that $\mathbb{1}[\vec{x}_k^{(i)} \neq \vec{x}_k^{(j)}] \sim \text{Bernoulli}(\frac{1}{2})$ independently. Therefore, by Chernoff's bound,

$$\begin{aligned}\Pr \left[\left\| \vec{x}^{(i)} - \vec{x}^{(j)} \right\| < \sqrt{d} \right] &= \Pr \left[\sum_{k=1}^d \mathbb{1}[\vec{x}_k^{(i)} \neq \vec{x}_k^{(j)}] < d/4 \right] \\ &= \Pr \left[\frac{d}{2} - \sum_{k=1}^d \mathbb{1}[\vec{x}_k^{(i)} \neq \vec{x}_k^{(j)}] \geq d/4 \right] \leq \exp(-d/8).\end{aligned}$$

By the union bound,

$$\Pr \left[\forall i \neq j : \left\| \vec{x}^{(i)} - \vec{x}^{(j)} \right\| \geq \sqrt{d} \right] \geq 1 - \binom{n}{2} \exp(-d/8) \geq 1 - \delta,$$

since $n \leq \sqrt{2\delta} \exp(d/16)$.

Exercise 2

(1) Since $f_{Z_i}(z) = f_{Z_i}(-z)$ for any $z \in \mathbb{R}$, $\Pr[X_i \geq z] = \Pr[X_i \leq -z]$ for $z \in \mathbb{R}$. Hence X_i has the same distribution as $-X_i$ and by independence, the empirical average A has the same distribution as $-A$. Therefore, $\mathbb{E}[X_i] = \frac{1}{2}(\mathbb{E}[X_i] + \mathbb{E}[-X_i]) = 0$ and $\Pr[|A| \geq \varepsilon] = \Pr[A \geq \varepsilon] + \Pr[-A \geq \varepsilon] = 2\Pr[A \geq \varepsilon]$.

(2) By Hoeffding's inequality, we have that

$$\Pr[A \geq \varepsilon] = \Pr[A - \mathbb{E}[A] \geq \varepsilon] \leq \exp\left(-\frac{n\varepsilon^2}{2M^2}\right).$$

(3) The following holds

$$\mathbb{E}[X_i^2] = 2 \int_0^M t^2 \frac{\exp(-t)}{2} dt = -(t^2 + 2t + 2)e^{-t} \Big|_0^M = 2 - (M^2 + 2M + 2)e^{-M}.$$

(4) By Bernstein's Inequality, we have that

$$\Pr[A \geq \varepsilon] = \Pr[A - \mathbb{E}[A] \geq \varepsilon] \leq \exp\left(-\frac{n\varepsilon^2/2}{\frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i^2] + \frac{1}{3}M\varepsilon}\right) \leq \exp\left(-\frac{n\varepsilon^2}{4 + 2M\varepsilon/3}\right),$$

where the last inequality follows from $\mathbb{E}[X_i^2] \leq 2$ for $i \in [n]$.

(5) Since $\varepsilon \leq M$ and $M \geq 2$, we have that

$$\exp\left(-\frac{n\varepsilon^2}{4+2M\varepsilon/3}\right) \leq \exp\left(-\frac{n\varepsilon^2}{M^2+2M\varepsilon/3}\right) \leq \exp\left(-\frac{n\varepsilon^2}{2M^2}\right).$$

Therefore, Bernstein's bound from Part (4) is tighter than the Hoeffding bound from Part (2).

(6) We have that

$$\begin{aligned} \mathbb{E}[\exp(tX_i)] &= \Pr[|Z_i| > M] + \int_0^M \frac{e^{tx} + e^{-tx}}{2} \exp(-x) dx \\ &= e^{-M} + \frac{1}{2} \left(\frac{1 - e^{(t-1)M}}{1-t} + \frac{1 - e^{-(t+1)M}}{t+1} \right) \\ &\leq e^{-M} + \frac{1}{2} \left(\frac{1}{1-t} + \frac{1}{t+1} - 2e^{-M} \right) = \frac{1}{1-t^2}. \end{aligned}$$

The inequality follows from the hint with $x = e^{-M}$.

(7) The following holds for $\varepsilon > 0$ and $t \in (0, 1)$,

$$\begin{aligned} \Pr[A \geq \varepsilon] &= \Pr[\exp(tA) \geq \exp(t\varepsilon)] \\ &\leq \frac{\mathbb{E}[\exp(tA)]}{\exp(t\varepsilon)} = \frac{\prod_{i=1}^n \mathbb{E}[\exp(tX_i/n)]}{\exp(t\varepsilon)} \quad (\text{Markov's ineq.}) \\ &\leq \frac{(1 - (t/n)^2)^{-n}}{\exp(t\varepsilon)} = \exp\left(-n \left(\ln(1 - \frac{t^2}{n^2}) + \frac{t\varepsilon}{n} \right)\right). \quad (\text{Part (6)}) \end{aligned}$$

Plugging $t = n(\sqrt{1 + \varepsilon^2} - 1)/\varepsilon$ into the above inequality, we complete the proof.

(8) Set $s = \sqrt{1 + \varepsilon^2} - 1$ and it is sufficient to show that

$$f(\varepsilon) = \ln \frac{2}{s+2} + s \geq \frac{2s(s+2)}{8 + s(s+2)}.$$

Let $g(x) = \ln(2/(x+2)) + x - \frac{2x(x+2)}{8+x(x+2)}$. We have that for $x \geq 0$,

$$g'(x) = \frac{x^2(x+1)(x^2+4x+20)}{(x+2)(x^2+2x+8)^2} \geq 0.$$

Therefore g is nondecreasing on $x \geq 0$ and hence, $g(x) \geq g(0) = 0$. Since

$$\exp(-nf(\varepsilon)) \leq \exp\left(-\frac{n\varepsilon^2}{4 + \frac{1}{2}\varepsilon^2}\right) \leq \exp\left(-\frac{n\varepsilon^2}{4 + 2\varepsilon^2/3}\right) \leq \exp\left(-\frac{n\varepsilon^2}{4 + 2M\varepsilon/3}\right),$$

the bound from Part (7) is tighter than Bernstein's inequality.

Exercise 3

(1) Fix the values of $S_1, \dots, S_{i-1}$. Let $Y := S_i - S_{i-1} \mid S_1, \dots, S_{i-1}$ and $Z := (Y + c)/2c$. We have that $\mathbb{E}[Z] = \frac{1}{2}$ and $Z \in [0, 1]$ a.s. Therefore,

$$\mathbb{E}[\exp(tY)] = e^{-ct} \mathbb{E}[\exp(2tcZ)] \leq e^{-ct} \mathbb{E}[1 - Z + e^{2tc}Z] = \frac{e^{tc} + e^{-tc}}{2},$$

where the inequality follows from the convexity of $x \mapsto \exp(x)$. Since

$$\frac{e^x + e^{-x}}{2} = \sum_{i=0}^{\infty} \frac{x^{2i}}{(2i)!} \leq \sum_{i=0}^{\infty} \frac{x^{2i}}{2^i i!} = e^{x^2/2},$$

$$\mathbb{E}[\exp(tY)] \leq \frac{e^{tc} + e^{-tc}}{2} \leq \exp(t^2 c^2 / 2).$$

(2) By Part (1), we have that

$$\mathbb{E}[\exp(t(S_i - S_{i-1})) \mid S_1, \dots, S_{i-1}] = \mathbb{E}[\exp(tS_i) \mid S_1, \dots, S_{i-1}] \exp(-tS_{i-1}) \leq \exp\left(\frac{t^2 c^2}{2}\right).$$

Hence, $\mathbb{E}[\exp(tS_i) \mid S_1, \dots, S_{i-1}] \leq \exp(\frac{t^2 c^2}{2} + tS_{i-1})$. Taking expectation on both sides, we conclude that

$$\begin{aligned} \mathbb{E}[\exp(tS_i)] &= \mathbb{E}[\mathbb{E}[\exp(tS_i) \mid S_1, \dots, S_{i-1}]] \\ &\leq \mathbb{E}\left[\exp\left(\frac{t^2 c^2}{2} + tS_{i-1}\right)\right] = \exp\left(\frac{t^2 c^2}{2}\right) \mathbb{E}[\exp(tS_{i-1})]. \end{aligned}$$

(3) By Part (2), we have that

$$\mathbb{E}[\exp(tS_n)] \leq \exp\left(\frac{t^2 c^2}{2}\right) \mathbb{E}[\exp(tS_{n-1})] \leq \dots \leq \exp\left(\frac{t^2 c^2}{2}\right)^n \mathbb{E}[\exp(tS_0)] = \exp\left(\frac{nt^2 c^2}{2}\right). \quad (1)$$

By Markov's inequality and Equation (1), for any $t > 0$, the following holds,

$$\Pr[S_n \geq \varepsilon n] = \Pr[\exp(tS_n) \geq \exp(t\varepsilon n)] \leq \frac{\mathbb{E}[\exp(tS_n)]}{\exp(t\varepsilon n)} \leq \exp\left(\frac{nt^2 c^2}{2} - t\varepsilon n\right). \quad (2)$$

Plugging $t = \frac{\varepsilon}{c^2}$ into Equation (2), we conclude that

$$\Pr[S_n \geq \varepsilon n] \leq \exp\left(-\frac{n\varepsilon^2}{2c^2}\right).$$

Exercise 4

(1) Since the samples in S are i.i.d., Bernstein's inequality and the union bound give

$$\begin{aligned} \Pr\left[\exists h \in H : L(h) - \hat{L}_S(h) \geq \varepsilon\right] &\leq |H| \max_{h \in H} \exp\left(-\frac{3n\varepsilon^2}{6L(h)(1-L(h)) + 2\varepsilon}\right) \\ &\leq |H| \exp\left(-\frac{3n\varepsilon^2}{6p(1-p) + 2\varepsilon}\right), \end{aligned}$$

since $p \leq 1/2$.

(2) Set

$$\varepsilon = \sqrt{\frac{2p(1-p) \ln(|H|/\delta)}{n}} + \frac{2}{3n} \ln(|H|/\delta).$$

By Part (1), it is sufficient to show that

$$|H| \exp\left(-\frac{3n\varepsilon^2}{6p(1-p) + 2\varepsilon}\right) \leq \delta \iff 3n\varepsilon^2 - 2 \ln\left(\frac{|H|}{\delta}\right) \varepsilon - 6 \ln\left(\frac{|H|}{\delta}\right) p(1-p) \geq 0. \quad (3)$$

By solving the quadratic inequality, Equation (3) holds if

$$\varepsilon \geq \frac{\ln(|H|/\delta)}{3n} + \sqrt{\frac{(\ln(|H|/\delta))^2}{9n^2} + \frac{2p(1-p)\ln(|H|/\delta)}{n}}. \quad (4)$$

Equation (4) immediately follows from $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ for $a, b \geq 0$.

Exercise 5

(1) By minimality, $L(\hat{h}_S) - L(h^*) \geq 0$. Then Part (1) follows from Markov's inequality.

(2) Let A be the event that $L(\hat{h}_S) - L(h^*) \geq \varepsilon$. Since $L(\hat{h}_S) - L(h^*) \leq L(\hat{h}_S) \leq 1$, we have that

$$\begin{aligned} \mathbb{E}[L(\hat{h}_S) - L(h^*)] &= \mathbb{E}[L(\hat{h}_S) - L(h^*) \mid \bar{A}] \Pr[\bar{A}] + \mathbb{E}[L(\hat{h}_S) - L(h^*) \mid A] \Pr[A] \\ &\leq \mathbb{E}[L(\hat{h}_S) - L(h^*) \mid \bar{A}] + \Pr[A] \leq \varepsilon + \delta. \end{aligned}$$

(3) Since $\mathbb{E}[X] = \int_0^\infty \Pr[X > x] dx$, we have that for any $\varepsilon > 0$,

$$\begin{aligned} E &= \int_0^\varepsilon \Pr[L(\hat{h}_S) - L(h^*) > x] dx + \int_\varepsilon^\infty \Pr[L(\hat{h}_S) - L(h^*) > x] dx \\ &\leq \int_0^\varepsilon 1 dx + \int_\varepsilon^\infty 2|H| \exp(-nx^2/2) dx \\ &= \varepsilon + \frac{2|H|}{\sqrt{n}} \int_{\sqrt{n}\varepsilon}^\infty \exp(-t^2/2) dt \\ &\leq \varepsilon + \frac{2|H|}{n\varepsilon} \exp(-n\varepsilon^2/2). \end{aligned} \quad (5)$$

Plugging $\varepsilon = \sqrt{\frac{2\ln(2|H|)}{n}}$ into Equation (5), we conclude that

$$E \leq \sqrt{\frac{2\ln(2|H|)}{n}} + \sqrt{\frac{1}{2n\ln(2|H|)}} = \sqrt{\frac{2\ln(2|H|)}{n}} \left(1 + \frac{1}{2\ln(2|H|)}\right).$$